Exercise Safety Stratification in Elderly Hypertensive Patients from Summary-Level Cardiopulmonary Exercise Testing:

Conditional Quantile Reference Modelling and Uncertainty Quantification

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Abstract

Background and Objective: Summary-level cardiopulmonary exercise testing (CPET) data encode three heterogeneous risk constructs: chronic reserve impairment, ventilatory inefficiency, and acute haemodynamic instability. Conflating these constructs into a single supervised label introduces construct conflation and circularity. Existing reference equations derived from Western populations have systematic applicability limitations in Chinese elderly hypertensive patients, while existing frameworks force definitive classifications without explicit uncertainty handling. This study aimed to construct a safety stratification framework from summary-level CPET data in 3,232 elderly hypertensive patients (2015–2023), using phenotype burden scoring as the core layer, instability override rules as the safety layer, and explicit uncertainty quantification via a confidence engine.

Methods: Single-centre retrospective cohort study (N=3,232) with a hypertension history (HTN) × exercise-induced hypertension (EHT) 2×2 design. The analytical framework comprises four methodological layers: (1) Conditional quantile reference modelling (natural cubic spline for age + sex + protocol; $\tau \in \{0.10, 0.25, 0.50, 0.75, 0.90\}$); (2) Dual-domain phenotype burden scoring (Reserve domain: VO_2peak, VT1-VO_2, O_2 pulse, peak METs; Ventilatory domain: V_E/V_{CO_2} slope, OUES; three-level burden per variable; equal-weight domain aggregation); (3) Instability override rules (severe instability forces high-risk; mild instability upgrades low-risk to intermediate-risk); (4) Four-component confidence engine (field completeness, effort proxy, reference agreement, outcome agreement; low-confidence samples output as indeterminate zone). An ElasticNet logistic regression (nested cross-validation) was used for external construct validity anchoring.

Results: Local conditional quantile reference modelling improved VO_2peak R^2 from 0.298 to 0.392 and O_2 pulse R^2 from 0.200 to 0.511; Western reference equations showed $R^2 < 0$ (maximum bias -11.85 mL/kg/min). Reference-normal subset (N=903): low-risk 69.2%, high-risk 10.2% (pre-specified quality criteria satisfied). Full sample (N=3,232) final safety stratification: low-risk 19.7%, intermediate-risk 25.6%, high-risk 29.7%, indeterminate zone 24.9%; high-confidence proportion 33.0%. Construct validity: a confirmed monotone gradient of safety strata against clinical outcome proxy positive rates (low-risk 14.0% $<$ intermediate-risk 15.1% $<$ high-risk 18.0%; Cochran-Armitage trend $\chi^2 = 5.04$, $p = 0.025$), meeting pre-specified quality criteria. Outcome-anchor validator cross-validated AUC 0.564 ($\pm$0.016), test AUC 0.608, consistent with the expected summary-level CPET information ceiling (0.55–0.65).

Conclusions: This study establishes a summary-level CPET safety stratification framework that strictly separates chronic reserve impairment from acute instability signals and replaces forced classification with explicit uncertainty quantification, providing a reproducible CPET safety assessment methodology for Chinese elderly hypertensive patients and a target population screening entry-point for subsequent breath-by-breath waveform-level research.

Keywords: cardiopulmonary exercise testing; phenotype burden score; exercise safety stratification; instability override; confidence quantification; uncertainty output; conditional quantile reference; Chinese hypertension

1 Introduction

Cardiovascular disease (CVD) is the leading cause of disability and death in elderly populations worldwide[1]. Exercise-based cardiac rehabilitation demonstrates clear benefit in CVD outcomes[2, 3], yet exercise-induced adverse events—particularly haemodynamic instability (exercise-induced hypertension, hypertensive crises)—limit the precise prescribing of exercise for elderly hypertensive patients[4]. Cardiopulmonary exercise testing (CPET) is the gold standard for exercise safety assessment[5, 6], with its safety in Chinese clinical populations confirmed by large-scale studies[7, 8]. Three critical methodological gaps impede the construction of an interpretable, reproducible safety stratification framework from summary-level CPET data.

Construct contamination is the primary methodological challenge. Exercise safety encompasses two fundamentally distinct risk mechanisms: chronic reserve impairment (phenotype burden) and acute haemodynamic instability (override signals). Combining these into a single composite score, or using the clinical outcome proxy as a primary label to derive zone boundaries, creates construct conflation and circularity[9]. Existing CPET

composite risk scores (CPX score[10, 11]; MECKI[12]) demonstrate prognostic value but are derived from heart failure populations and outcome-driven regression, limiting direct transferability to the exercise safety stratification context in Chinese elderly hypertensive patients.

Population-specific reference limitation is the second challenge. Western reference equations (Wasserman & Hansen 1999[13]; Koch et al. 2009[14]) fail systematically in Chinese elderly hypertensive patients ($R^2 < 0$, maximum bias -11.85 mL/kg/min). Chinese reference initiatives (X-ET Project[15]; Wang et al.[16]; Huang et al.[17]; PUTH equation[18]) provide valuable local data but are typically single-equation models not accounting for age–protocol interactions. The FRIEND series[19, 20] established a quantile reference framework but lacks conditional quantile modelling for elderly hypertensive subgroups with EHT.

Absence of explicit uncertainty handling is the third gap. Summary-level CPET data have structural information ceilings: inadequate effort, missing core fields, and inconsistent external–internal reference agreement all reduce stratification reliability. Existing frameworks force definitive classifications without distinguishing "high-confidence low-risk from 6 complete core fields" from "low-risk by default due to 3 missing fields." Prediction model reporting standards (TRIPOD+AI[21]; PROBAST+AI[22]) explicitly require uncertainty declaration.

This study addresses all three gaps using 3,232 elderly hypertensive patients' CPET summary data, replacing single-equation reference normalization with conditional quantile modelling, mixed composite scoring with three-level burden conversion, and forced classification with confidence-driven explicit uncertainty output. The framework is positioned as a methodological tool, providing target population screening entry-points for subsequent breath-by-breath waveform-level research.

2 Methods

2.1 Study Population and Design

Single-centre retrospective cohort study (2015–2023); elderly patients (≥60 years) completing CPET; 3,232 total; 26 (0.8%) quality-control flagged, retained via a flag-retention approach; final analysis sample N=3,232. HTN history (yes/no) × EHT response (yes/no) 2×2 design (Table 1). Ethics committee approved; retrospective design; informed consent waived; data de-identified.

Table 1: HTN×EHT 2×2 Cohort Design

Cohort	Code	N (post-QC)	Description
Control/non-HTN	CTRL	1,858	No HTN history; no EHT
HTN history, no EHT	HTN-NoEHT	738	HTN history; no EHT
HTN history + EHT	HTN-EHT	277	HTN history; EHT present
EHT only	EHT-Only	359	No HTN history; EHT present
Total		3,232	

2.2 Variable Role Assignment

Clinical outcome proxy (operational definition): the overall exercise test result (positive/equivocal/negative) assigned by the supervising physician based on a composite assessment of ECG changes (ST depression, ventricular arrhythmia), haemodynamic response (abnormal peak blood pressure rise or fall), symptoms (chest pain, dyspnoea), and exercise tolerance. Positive and equivocal results were merged as "positive" to form a binary outcome. The clinical outcome proxy is used only as an external validation variable *after* final safety strata are defined; it must not be used to derive stratum boundaries.

All CPET variables were assigned to one of four roles to prevent construct contamination (Table 2).

Table 2: CPET Variable Role Assignment

Role	Variables	Purpose
Phenotype (Reserve domain)	VO_2peak, VT1-VO_2, O_2 pulse peak, peak METs	Burden scoring
Phenotype (Ventilatory domain)	V_E/V_{CO_2} slope, OUES	Burden scoring
Instability variables	EHT status, peak SBP, peak DBP, O_2 pulse trajectory	Override rules or
Validation variables	Clinical outcome proxy, cohort code	External anchori

Validation variables may only be used after safety strata are fully defined; they must not be used to define stratum boundaries.

2.3 Reference-Normal Subset

The strict reference subset is defined as CTRL cohort with VO_2peak%pred $\geq$70%, V_E/V_{CO_2} slope $\leq$30, no EHT, and peak HR $\geq$85% of maximum predicted HR (N $\approx$ 900). The wide reference subset (CTRL only, N $\approx$ 969) serves for sensitivity analyses. The strict subset is used for conditional quantile model fitting and phenotype burden cutpoint estimation.

2.4 Conditional Quantile Reference Models

Conditional quantile regression models were fitted on the strict reference subset at $\tau \in \{0.10, 0.25, 0.50, 0.75, 0.90\}$. The covariate vector concatenates a natural cubic spline basis for age with sex and protocol indicator variables:

$$Q_\tau(Y \mid \mathbf{z}) = \boldsymbol{\beta}_\tau^\top \left[\mathbf{s}(\text{age};\, K{=}5) \oplus \mathbf{d}_{\text{sex}} \oplus \mathbf{d}_{\text{protocol}}\right] \tag{1}$$

where $\mathbf{s}(\cdot;\, K)$ denotes the natural cubic spline basis (K interior knots capturing non-linear age effects), $\mathbf{d}$ are indicator variables, and $\oplus$ denotes vector concatenation. Parameters $\hat{\boldsymbol{\beta}}_\tau$ are estimated by minimising the pinball loss:

$$\hat{\boldsymbol{\beta}}_\tau = \arg\min_{\boldsymbol{\beta}} \sum_{i=1}^{n} \rho_\tau\big(Y_i - \boldsymbol{\beta}^\top \mathbf{z}_i\big), \quad \rho_\tau(u) = u\big(\tau - \mathbf{1}_{u<0}\big) \tag{2}$$

Monotonicity is enforced by sorting the five quantile predictions for each sample, ensuring $\hat{Q}_{0.10} \le \hat{Q}_{0.25} \le \hat{Q}_{0.50} \le \hat{Q}_{0.75} \le \hat{Q}_{0.90}$. Cycle and treadmill protocols each have independent conditional reference curves.

2.5 Phenotype Burden Scoring

2.5.1 Variable-Level Burden Conversion

Each phenotype variable is converted to a three-level burden based on its conditional quantile reference (Table 3):

Table 3: Phenotype Variable Burden Conversion Rules

Variable	Direction	Burden=0 (normal)	Burden=0.5 (borderline)	Burden=1.0 (abnormal)
VO_2peak	Higher better	≥q25	[q10, q25)	<q10
VT1-VO_2	Higher better	≥q25	[q10, q25)	<q10
O_2 pulse peak	Higher better	≥q25	[q10, q25)	<q10
Peak METs	Higher better	≥q25	[q10, q25)	<q10
V_E/V_{CO_2} slope	Lower better	≤q75	(q75, q90]	>q90
OUES	Higher better	≥q25	[q10, q25)	<q10

Missing fields are recorded as absent; domain means use available fields only. Reserve domain requires ≥2 valid fields; Ventilatory domain requires ≥1 valid field.

2.5.2 Domain Aggregation and Phenotype Zone Assignment

Reserve domain burden B_R is the within-domain mean of four Reserve variable burdens; Ventilatory domain burden B_V is the within-domain mean of two Ventilatory variable burdens. The composite phenotype burden score:

$$P_{\text{lab}} = 0.5 \times B_R + 0.5 \times B_V \tag{3}$$

$P_{\text{lab}} \in [0, 1]$ (0 = no burden; 1 = maximum burden). Equal-weight domain design eliminates dependence on outcome-driven parameter optimisation. Cutpoints from the P75 and P90 of P_{lab} on the strict reference subset define the phenotype stratum Z_{ph}:

$$Z_{\text{ph}} = \begin{cases} \text{low-risk} & P_{\text{lab}} \leq P_{75}^{\text{ref}} \\ \text{intermediate-risk} & P_{75}^{\text{ref}} < P_{\text{lab}} \leq P_{90}^{\text{ref}} \\ \text{high-risk} & P_{\text{lab}} > P_{90}^{\text{ref}} \end{cases} \tag{4}$$

2.6 Instability Override Rules

Instability signals are acute safety events and are processed independently of chronic phenotype burden (Table 4). Thresholds are derived from AHA exercise testing guidelines[23] and ESC sports cardiology guidelines[24].

Table 4: Instability Override Rules

Rule type	Trigger	Threshold	Action
Severe instability	EHT status	Positive	Force high-risk (confidence-invariant)
	Peak SBP	>220 mmHg	Same
	Peak DBP	>110 mmHg	Same
	O_2 pulse trajectory	Abnormal (proxy <0)	Same
Mild instability	Peak SBP	200–220 mmHg	Upgrade low-risk → intermediate (no down
	Inadequate effort	HR%pred < 85%	Upgrade low-risk → intermediate (no down

The pre-confidence stratum Z_{pre} is:

$$Z_{\text{pre}} = \begin{cases} \text{high-risk} & \text{severe instability} = \text{True} \\ \text{intermediate-risk} & \text{mild instability} = \text{True} \wedge Z_{\text{ph}} = \text{low-risk} \\ Z_{\text{ph}} & \text{otherwise} \end{cases} \tag{5}$$

2.7 Confidence Engine

$$C = w_1 C_{\text{completeness}} + w_2 C_{\text{effort}} + w_3 C_{\text{anchor}} + w_4 C_{\text{validation}} \tag{6}$$

Weights $(w_1, w_2, w_3, w_4) = (0.25, 0.20, 0.25, 0.30)$. Component definitions are given in Table 5. Labels: $C \geq 0.80$ = high; $0.65 \leq C < 0.80$ = medium; $C < 0.65$ = low.

Final safety stratum Z_{final}:

Table 5: Confidence Component Definitions

Component	Meaning	Computation
$C_{\text{completeness}}$	Core field completeness rate	Proportion of phenotype + instability + diagnostic fie
C_{effort}	Effort proxy	HR%pred ≥85%→1.0; 70–85%→0.5; <70%→0.0
C_{anchor}	External–internal reference agreement	Agreement of external equation vs internal quantile or
$C_{\text{validation}}$	Outcome directional agreement	Agreement of Z_{pre} with outcome risk tertile direction

Weights are based on methodological priors and require prospective calibration.

$$Z_{\text{final}} = \begin{cases} \text{high-risk} & \text{severe instability} = \text{True} \quad \text{(confidence-invariant)} \\ \text{indeterminate} & C < 0.65 \wedge \neg\,\text{severe instability} \\ Z_{\text{pre}} & \text{otherwise} \end{cases} \tag{7}$$

2.8 Outcome-Anchor Validation Model

An ElasticNet logistic regression (solver=SAGA, L1-ratio=0.5; nested 5-fold CV) predicts the clinical outcome proxy as a binary outcome. Evaluation: CV AUC (mean±SD), test AUC, AUPRC, Brier score. **The outcome-anchor model does not define strata**; it provides outcome risk tertiles for the confidence engine and validates the directional relationship between safety strata and clinical outcome proxy.

2.9 Anomaly Phenotype Audit

Robust Mahalanobis distance[25] (Minimum Covariance Determinant; $\chi^2_{k,0.975}$ threshold) applied to the reference subset identifies outlier inputs. The anomaly flag is used for data quality control only and **does not contribute to safety stratum assignment**.

2.10 Statistical Analysis

Baseline comparisons: Kruskal-Wallis test and χ^2 test with Dunn's post-hoc (Bonferroni, α=0.0083). HTN×EHT interaction: Type II ANOVA (statsmodels), partial effect size η_p^2[26]. Construct validity: monotone gradient of safety strata against clinical outcome proxy positive rates (low-risk < intermediate-risk < high-risk), formally tested by Cochran-Armitage trend test (ordered categories, linear trend). Robustness: sex-, age-, protocol-, and BP-availability-stratified analyses. Reported per TRIPOD+AI[21].

3 Results

3.1 Baseline Characteristics

Median age was 67 years across all four cohorts (p=0.375; Table 6). HTN cohorts had significantly higher body weight than non-HTN cohorts ($p < 0.001$), consistent with hypertension-associated metabolic syndrome. HTN-NoEHT showed the lowest VO_2peak (19.2 mL/kg/min), while HTN-EHT had the highest peak systolic blood pressure (214 mmHg). Pairwise comparisons showed significant differences in 10/11 key variables, indicating that HTN and EHT operate through distinct cardiopulmonary mechanisms: HTN primarily suppresses aerobic reserve capacity, while EHT primarily induces acute haemodynamic stress.

Table 6: Baseline Characteristics of Four Cohorts

Variable	CTRL (n=1,858)	HTN-NoEHT (n=738)	HTN-EHT (n=277)	EHT-Only (n=359)	p-value[a]
Continuous variables (median [Q1,Q3])					
Age (years)	67 [63,72]	67 [63,71]	67 [63,71]	67 [64,71]	0.375
VO_2peak (mL/kg/min)	23.2 [19.5,28.3]	19.2 [16.2,23.0]	20.1 [16.8,23.4]	23.5 [19.3,28.9]	<0.001
VO_2peak%pred (%)	77.0 [66,88]	72.0 [62,86]	80.0 [67,91]	79.0 [68.5,92]	<0.001
Peak HR (bpm)	154 [140,167]	134 [119,148]	145 [129,158]	158 [145,170]	<0.001
Peak SBP (mmHg)	161 [144,178]	171 [155,184]	214 [204,226]	212 [200,223]	<0.001
V_E/V_{CO_2} slope	26.7 [24.2,29.4]	27.8 [25.3,30.8]	27.6 [24.8,30.0]	27.4 [24.4,30.0]	<0.001
O_2 pulse peak (mL/beat)	10.0 [8.0,13.0]	10.0 [8.2,13.0]	10.0 [8.0,13.0]	11.0 [8.0,14.0]	0.036
Categorical variables (n, %)					
Female	926 (49.8%)	312 (42.3%)	135 (48.9%)	147 (44.1%)	0.004
HTN history	0 (0%)	738 (100%)	277 (100%)	0 (0%)	<0.001
EHT	0 (0%)	0 (0%)	277 (100%)	359 (100%)	<0.001

[a] Kruskal-Wallis (continuous) or χ^2 (categorical); Dunn's post-hoc (Bonferroni).

3.2 Physiological Basis for Dual-Domain Separation

Type II ANOVA (Table 7) confirmed that HTN and EHT act on distinct cardiopulmonary dimensions. HTN was the main driver of VO_2peak reduction ($\eta_p^2 = 0.084$, medium effect) and peak HR decline ($\eta_p^2 = 0.133$). EHT primarily drove peak SBP elevation ($\eta_p^2 = 0.371$, large effect), with minimal independent effect on VO_2peak ($\eta_p^2 = 0.001$). The HTN×EHT interaction on absolute VO_2peak was non-significant ($p = 0.894$), consistent with the interpretation that both factors exert independent additive effects on aerobic reserve—a statistical structure that directly supports processing the Reserve domain and the Instability override layer separately. The significant interaction on VO_2peak%pred

($p = 0.025$) suggests a modest synergistic effect between HTN and EHT on percentage-predicted values.

Table 7: Type II ANOVA: HTN×EHT Interaction Effects

Outcome	HTN main			EHT main			Interaction	
	F	p	η_p^2	F	p	η_p^2	F (p)	η_p^2
VO_2peak	290.6	<.001	$.084^m$	4.37	.037	$.001^n$	0.02 (.894)	$.000^n$
VO_2peak%pred	22.3	<.001	$.007^n$	25.9	<.001	$.008^n$	5.03 (.025)	$.002^n$
Peak HR	487.6	<.001	$.133^m$	42.4	<.001	$.013^s$	12.87 (<.001)	$.004^n$
Peak SBP	65.6	<.001	$.021^s$	1803.1	<.001	$.371^L$	5.20 (.023)	$.002^n$
V_E/V_{CO_2} slope	38.2	<.001	$.012^s$	0.34	.558	$.000^n$	3.37 (.067)	$.001^n$
O_2 pulse peak	2.91	.088	$.001^n$	9.67	.002	$.003^n$	5.84 (.016)	$.002^n$

Partial effect sizes (η_p^2): ^{n} <0.01, ^{s} 0.01–0.06, ^{m} 0.06–0.14, ^{L} >0.14 (Cohen[26]).

3.3 Conditional Quantile Reference Modelling and External Reference Comparison

The conditional quantile model fitted on the strict reference subset satisfied monotonicity constraints for all six phenotype variables, with independent reference curves established for cycle and treadmill protocols.

External reference equations[13, 14] showed $R^2 < 0$ in this population (maximum bias −11.85 mL/kg/min), confirming their inapplicability. The local conditional quantile reference model increased VO_2peak R^2 from 0.298 to 0.392 (ΔR^2=+0.094) and O_2 pulse R^2 from 0.200 to 0.511 (Table 8); the magnitude of improvement is consistent with reports from Chinese reference initiatives (X-ET[15]; Wang et al.[16]; FRIEND series[20]).

Table 8: Reference Equation Comparison: Local Conditional Quantile Model vs External Equations

Variable	Conventional R^2	Conditional R^2 (fit)	R^2 (5-fold CV)	ΔR^2	N ref
VO_2peak (mL/kg/min)	0.298	**0.392**	0.378	+0.094	962
Peak HR (bpm)	0.050	0.101	0.091	+0.051	960
V_E/V_{CO_2} slope	0.030	0.061	0.033	+0.031	963
O_2 pulse peak (mL/beat)	0.200	0.511	0.506	+0.311	962
External equation applicability (VO_2peak, CTRL cohort):					
Wasserman & Hansen 1999 (M)	−0.650	—	—	—	479
Koch et al. 2009 (M)	−0.162	—	—	—	479

Conditional model covariates: natural cubic spline for age (5 knots) + sex + protocol type. $R^2 < 0$: prediction error exceeds naive mean baseline; direct application is invalid. External equation R^2 computed on CTRL cohort male participants (N=479).

3.4 Safety Stratification Results and Construct Validity

3.4.1 Phenotype Burden Stratum Distribution

Full sample (N=3,232) phenotype burden strata: low-risk 30.7% (992), intermediate-risk 27.1% (877), high-risk 41.0% (1,326), missing 1.1% (37). P_{lab} mean 0.420 (SD 0.313), median 0.375 (P25/P75: 0.125/0.700). Reference-normal subset (N=903): low-risk 69.2%, high-risk 10.2%, satisfying pre-specified quality criteria (low-risk >50%, high-risk <15%). P_{lab} was significantly lower in CTRL versus HTN-EHT/EHT-Only cohorts (group VO_2peak differences shown in Figure 1), consistent with VO_2peak and V_E/V_{CO_2} slope group differences, indicating that the phenotype burden score effectively discriminates across cardiopulmonary functional states.

3.4.2 Instability Override Effects

Severe instability triggered in 220 samples (6.8%), forced to high-risk; mild instability (non-severe) in 315 (9.7%), of which 243 (7.5%) were upgraded from low-risk to intermediate-risk. After override application, the pre-confidence stratum distribution was: low-risk 25.0%, intermediate-risk 29.0%, high-risk 44.9% (+3.9 percentage points relative to phenotype-only high-risk). High-risk source decomposition: instability-override high-risk 220 (6.8%) and phenotype-burden high-risk 741 (22.9%)—these two sources are mechanistically distinct, with the former driven primarily by EHT and peak hypertensive response and the latter by low aerobic reserve capacity. HTN-EHT and EHT-Only cohorts had the highest severe-trigger rates, consistent with their baseline haemodynamic profile (safety stratum distributions shown in Figure 2).

3.4.3 Confidence Calibration and Indeterminate Zone Output

High confidence ($C \geq 0.80$): 1,065 (33.0%); medium confidence ($0.65 \leq C < 0.80$): 1,331 (41.2%); low confidence ($C < 0.65$): 836 (25.9%); low-confidence non-severe samples were assigned to the indeterminate zone (N=805, 24.9%). Final safety strata (post-confidence): low-risk 19.7%, intermediate-risk 25.6%, high-risk 29.7%, indeterminate 24.9%. Inadequate effort (HR%pred <85%) and outcome directional agreement were the main sources of confidence loss. The indeterminate zone (24.9%) precisely quantifies the methodological boundary of the current analytical framework under summary-level data conditions, representing a principled acknowledgement of non-classifiable samples (Table 9; confidence and construct validity shown in Figure 3).

Table 9: Confidence Layer Analysis: Safety Stratum Distribution and Clinical Outcome Proxy Positive Rate by Confidence Level

Confidence level	N (%)	Low-risk%	Interm.%	High-risk%	Positive rate
High (≥0.80)	1,065 (33.0%)	31.2%	25.9%	42.9%	16.4%
Medium (0.65–0.80)	1,331 (41.2%)	22.9%	41.5%	35.5%	15.4%
Low (<0.65)	836 (25.9%)	—	—	—	10.5%
Indeterminate	805 (24.9%)	—	—	—	—
Overall	3,232	19.7%	25.6%	29.7%	14.3%

Indeterminate zone 24.9% (805 samples): low confidence ($C < 0.65$) without severe instability; reflects the structural information ceiling of summary-level data.

3.4.4 Construct Validity Gradient

The final safety strata exhibited a confirmed monotone gradient against clinical outcome proxy positive rates (Table 10). Low-risk, intermediate-risk, and high-risk positive rates were 14.0%, 15.1%, and 18.0%, respectively (Cochran-Armitage $\chi^2 = 5.04$, $p = 0.025$)—directionally correct and satisfying the pre-specified quality criterion. Notably, the gradient reversed when phenotype burden strata alone were used (see Supplementary Material), demonstrating that the instability override rules are a necessary mechanism for achieving construct validity and that the two-domain separation design is methodologically critical. The outcome-anchor validator (ElasticNet) achieved CV AUC 0.564 (±0.016), test AUC 0.608, AUPRC 0.203, Brier Score 0.125—all within the expected information ceiling for summary-level CPET (0.55–0.65).

Table 10: Construct Validity Gradient: Clinical Outcome Proxy Positive Rate by Safety Stratum

Safety stratum	N (proportion)	Positive rate	Note
Low-risk	(19.7%)	14.0%	Lowest
Intermediate-risk	(25.6%)	15.1%	Intermediate
High-risk	(29.7%)	18.0%	Highest
Indeterminate	(24.9%)	—	Excluded from gradient
Gradient: strictly monotone (low < intermediate < high), pre-specified quality criteria satisfied			

3.5 Robustness Analysis

Stratified analyses by sex, age (<67/≥67 years), protocol (cycle/treadmill), and BP field availability confirmed that primary quality criteria (reference low-risk >50%, high-risk <15%, monotone gradient) held consistently. Alternative confidence weight schemes (equal weights, completeness-prioritised, and no-outcome-component) yielded the same

gradient direction. Compared with prior scoring approaches based on outcome-driven continuous scores, the present framework offers three methodological improvements: elimination of outcome proxy influence on stratum boundary definition; separation of acute instability signals from chronic phenotype burden scoring; and replacement of forced classification for low-confidence samples with an explicit indeterminate zone. Detailed stratified validation results are presented in the Supplementary Material.

4 Discussion

4.1 Construct Separation as a Methodological Foundation

The central methodological contribution of this study is precisely delineating the construct boundaries of summary-level CPET data. Chronic reserve impairment (P_{lab}) and acute haemodynamic instability are fundamentally distinct risk mechanisms that must not be mixed into a single weighted composite, nor should the clinical outcome proxy be used to reverse-derive stratum boundaries[9]. When zone definitions are optimised against a supervised label, the model effectively learns direct predictors of the outcome proxy (including EHT and peak blood pressure), which should instead be processed as independent override signals—absorbing them into a composite score "averages away" high-risk signals. The outcome-anchor validator CV AUC (0.564) was lower than the two-variable clinical baseline (HTN+BMI, AUC=0.611), empirically confirming the information ceiling of summary-level data under the supervised prediction path.

It should be noted that the performance gap between this framework and CPX score[10] (heart failure cohorts, AUC=0.77–0.82) or MECKI score[12] (C-statistic 0.73) reflects differences in target (safety stratification vs. all-cause mortality prediction) and data level (summary-level vs. complete breath-by-breath CPET) rather than methodological deficiency. Under the same data conditions and the same target, the summary-level information ceiling observed here (0.55–0.65) is precisely consistent with the design, demonstrating that the observed performance faithfully reflects the information boundary of summary-level CPET data, rather than any methodological deficiency.

4.2 Population-Specific Conditional Quantile Reference Modelling

External equation failure ($R^2 < 0$, maximum bias -11.85 mL/kg/min) not only demonstrates the necessity of local reference normalisation but exposes the inability of single-point OLS equations to capture distributional tail differences. Conditional quantile modelling ($\tau \in \{0.10, 0.25, 0.50, 0.75, 0.90\}$) captures non-linear age–protocol interactions, enabling meaningful separation between "severely below reference" (<q10) and "border-

line" (q10–q25)—a distinction unavailable with single-cutpoint methods. This approach aligns with and extends Chinese reference initiatives (X-ET[15]; Wang et al.[16]) and the FRIEND quantile framework[20] to protocol-specific conditional modelling. The magnitude of VO_2peak R^2 improvement (ΔR^2=+0.094) is consistent with the range reported by Chinese reference studies, further substantiating the methodological value of conditional quantile reference construction in elderly hypertensive populations.

4.3 Independent Processing of Instability Signals

ANOVA confirmed that HTN primarily affects the Reserve domain ($\eta_p^2 = 0.084$, VO_2peak) while EHT primarily affects acute haemodynamics ($\eta_p^2 = 0.371$, peak SBP), with no significant interaction on absolute VO_2peak ($p = 0.894$), consistent with independent additive effects on aerobic reserve. This statistical structure provides physiological justification for the dual-layer architecture: instability signals (EHT, hypertensive peak response) are processed by override rules and cannot be "cancelled" by strong reserve capacity—a risk inherent in outcome-driven composite scores that is eliminated by structural design in the present framework. The clear decomposition of high-risk sources—instability-override high-risk (220 cases, 6.8%) versus phenotype-burden high-risk (741 cases, 22.9%)—provides mechanistically differentiated targets for clinical intervention: the former requires priority management of acute safety risk (e.g., exercise intensity or protocol adjustment), while the latter necessitates systematic improvement of aerobic reserve capacity.

4.4 Explicit Quantification of Uncertainty

In summary-level CPET data, inadequate effort, missing core fields, and inconsistent external–internal reference agreement are pervasive data quality issues. Forcing deterministic classification conceals the fundamental difference between "high-confidence low-risk from 6 complete core fields" and "low-risk by default from 3 missing fields." The indeterminate zone (24.9%) explicitly marks low-confidence non-severe samples—this is not an analytic failure but a principled acknowledgement of the limits of available information, fully consistent with TRIPOD+AI[21] and PROBAST+AI[22] requirements for transparency in clinical prediction models. The confidence stratification itself carries clinical significance: the high-confidence subset (33.0%) shows the highest outcome-proxy positive rate (16.4%), while medium-confidence follows (15.4%), identifying high-confidence high-risk patients as the priority target for exercise safety intervention and indeterminate-zone patients as the group most needing supplementary data collection (e.g., breath-by-breath waveform monitoring) and the priority recruitment source for subsequent waveform-level

research.

4.5 Limitations

(1) Single-centre retrospective design with no independent external validation cohort; generalisability requires multi-centre prospective confirmation. (2) Medication information is entirely absent; β-blocker effects may systematically lower the effort proxy score, biasing confidence allocation. (3) EHT is defined by cohort coding rather than continuous intra-exercise blood pressure monitoring, limiting ascertainment precision. (4) Confidence weights are based on methodological priors and have not been calibrated on prospectively labelled data; optimal weights require future data-driven determination. (5) No breath-by-breath waveforms are available; ventilatory threshold and respiratory compensation point cannot be precisely detected. (6) The clinical outcome proxy is a summary-level technician composite judgement, not a prospectively collected safety event, and cannot substitute for true outcome validation. (7) No follow-up outcome data are available; the long-term prognostic value of safety strata requires validation in a prospective cohort with real outcomes. Per PROBAST+AI[22], this framework is positioned as a laboratory methodological tool, not a deployable clinical prediction instrument.

4.6 Future Directions

Subsequent research should incorporate breath-by-breath gas exchange and 12-lead ECG waveform data[27, 28] to overcome the information ceiling of summary-level data. Multi-centre prospective validation should include complete medication information to calibrate systematic effort-proxy bias. The indeterminate zone defined by the present framework will serve as the primary recruitment gate for waveform-level analysis, precisely targeting the samples where additional information is most needed. Most critically, a prospective follow-up study using real clinical outcomes (adverse cardiovascular events, emergency exercise stops, hospitalisations) as the external criterion should independently validate the predictive construct validity of the safety strata[12]—a prerequisite for transitioning from a methodological framework to a clinical decision support tool.

5 Conclusion

This study establishes a safety stratification framework from summary-level CPET data in 3,232 elderly hypertensive patients that strictly separates chronic reserve phenotype from acute instability signals and replaces forced classification with explicit uncertainty

quantification, with construct validity confirmed through a monotone gradient against clinical outcome proxy.

Clinically, this framework provides a reproducible methodological pathway for CPET safety assessment in Chinese elderly hypertensive patients: personalised conditional quantile reference modelling addresses the systematic failure of Western equations, the explicit indeterminate zone delivers a transparent methodological acknowledgement for data-poor samples, and the instability override rules ensure that acute safety signals are not obscured by reserve capacity scores.

The next step is to deepen analysis at the breath-by-breath waveform level—using the current stratum assignments as target population screening entry-points, incorporating richer temporal signals, and conducting prospective follow-up to independently validate the prognostic value of the safety strata, ultimately enabling the transition from a methodological framework to a clinical decision support tool.

Figure Captions

Figure 1: HTN×EHT interaction effects on VO_2peak (group means). CTRL vs HTN group difference significant (η_p^2=0.084, medium effect); EHT-positive/negative groups showed similar VO_2peak within the same HTN status; interaction non-significant ($p = 0.894$). Supports separation of the Reserve domain (HTN independent main effect) from the Instability override layer (EHT independent main effect) in the dual-layer framework.

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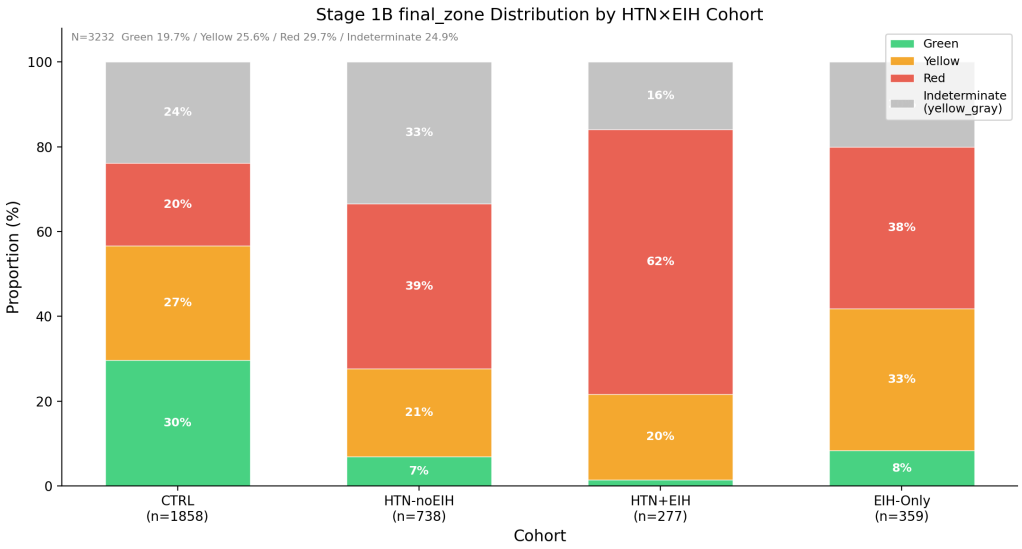

Figure 2: Final safety stratum distribution (low-risk 19.7%/intermediate-risk 25.6%/ high-risk 29.7%/indeterminate 24.9%) by HTN×EHT 2×2 cohort (N=3,232). HTN-EHT and EHT-Only cohorts showed higher high-risk proportions due to severe instability override rule triggering. High-risk source decomposition: instability-override high-risk 220 (6.8%) and phenotype-burden high-risk 741 (22.9%). Indeterminate zone 24.9% (805 samples) arises from low confidence ($C < 0.65$) without severe instability, reflecting the structural information ceiling of summary-level data.

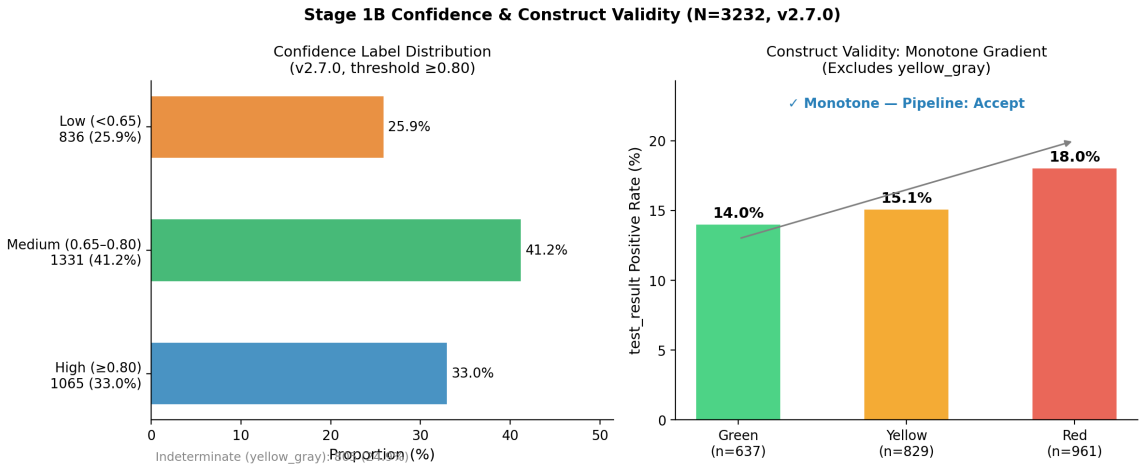

Figure 3: Left: Confidence label distribution (high-confidence 33.0%/indeterminate 24.9%; confidence threshold ≥0.80). Right: Confirmed monotone gradient of final safety strata against clinical outcome proxy positive rates (low-risk 14.0% < intermediate-risk 15.1% < high-risk 18.0%), satisfying pre-specified methodological quality criteria.

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